

Edward Donson

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EDUCATION AND RELEVANT COURSEWORK

Harvey Mudd College

Claremont, CA

BS - Mathematics and Computer Science — High Distinction, Honors in Both Fields

May 2026

- Data Structures & Algorithms
- Linear Algebra
- Intermediate Probability
- Operations Research
- Stochastic Processes
- Network Algorithmics

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, HTML/CSS, R, SQL

Frameworks: React, Node.js, Flutter, MySQL, MongoDB

Developer Tools: Git, VS Code, Eclipse, Linux/UNIX, Apache, Docker

Libraries: pandas, mongoose, NumPy, Matplotlib, ASE, AMP, PyTorch, scikit-learn, Express, rJAGS

EXPERIENCE

Operations Research Intern

Jan. 2026 – May 2026

Mudd OR Group, Harvey Mudd College

Claremont, CA

Principal Investigator: Dr. Susan Martonosi

- Develop interactive UI interface for Claremont College swim coaches to validate swim meet line-ups
- Implement Mixed-Integer Program based model to produce potential NCAA portfolio based on past swimmer performance

Software Engineer

Aug. 2025 – May 2026

Sokil Inc.

Claremont, CA

- Developing software to extract LiDAR and IMU data from proprietary file .lvx file type
- Leveraging eigenvalue-based curvature metrics to discern landmine presence in LiDAR scans, employing clustering and feature detection to highlight potential landmines and remove extraneous data points

Adversarial Machine Learning Research Intern

May. 2025 – Aug. 2025

Center for Artificial Intelligence Security Research, Oak Ridge National Laboratory

Oak Ridge, TN

Principal Investigator: Dr. Amir Sadovnik

- Developed extraction attack methods which leverage finetuned pre-trained vision models to produce transferable evasion attacks for a select target image
- Leveraged hessian analysis and loss surface visualization methods to theoretically validate extraction attack effectiveness, ensuring extraction attacks properly mimic target model loss surfaces

Data Science Research Intern

Jan. 2025 – Jan. 2026

HADD Research Group, Harvey Mudd College

Claremont, CA

Principal Investigator: Dr. Jamie Haddock

- Conduct experiments on the accuracy of sketch-solve methods for non-negative least squares regression
- Compare existing randomized non-negative least squares regression with sketch-and-solve methods using similarity metrics like Jaccard similarity

LLM Education Research Intern

Aug. 2024 – Aug. 2025

METRICS Lab, Harvey Mudd College

Claremont, CA

Principal Investigator: Dr. Zachary Dodds

- Interpret hard metrics, like cyclomatic complexity and comment-code percentage, to understand LLM's effect on student's code writing habits
- Developed a Python script to convert and filter .ipynb files to produce compilable python code for static analysis

Machine Learning Research Assistant

Jan. 2024 – Sep. 2024

MolSim Lab, Harvey Mudd College

Claremont, CA

Principal Investigator: Dr. Sandra Brown

- Developed universal scripts to analyze neural network performance using pandas and NumPy and produce visualizations illustrating neural networks accuracy

- Trained artificial neural networks to predict molecule's potential energy using the Atomistic Potential Learning package for three molecules of differing complexity

Computer Science Grader and Tutor

Aug. 2023 – Aug. 2025

Department of Computer Science, Harvey Mudd College

Claremont, CA

- Improved student performance and understanding of programming principles, data structures or logic and computability through tutoring sessions attended by up to 25 students
- Collaborated with other graders to provide feedback for classes of up to 200 students, resulting in 100% of assignments having comprehensive feedback and grades

PUBLICATIONS

- Yuan Garcia, Jenny Ngo, Florence Rui Lin, Edward Donson, and Zachary Dodds. 2025. Adaptable Metrics to Assess and Improve Introductory CS. Proceedings of the ACM Global Computing Education Conference 2025 Vol 1 (CompEd 2025), October 21-25, 2025, Gaborone, Botswana. doi: 10.1145/3736181.3747157 [**Best Student Paper Runner-Up**]

PRESENTATIONS

SPiE Defense + Security 2026 Conference | SPiE

April 2026

- Edward Donson and Amir Sadovnik. 2026. Query-Efficient Finetuning to Improve Adversarial Attack Transferability. SPiE, April 28, 2026, National Harbor, MD.

Science Undergraduate Laboratory Internship Symposium | Oak Ridge National Laboratory

July 2025

- Edward Donson and Amir Sadovnik. 2025. Query-Efficient Finetuning to Improve Adversarial Attack Transferability. ORNL, July 16, 2025, Oak Ridge, TN.

Consortium for Computer Sciences in Colleges Southwest | UC San Diego

Mar. 2025

- Yuan Garcia, Jenny Ngo, Florence Rui Lin, Edward Donson, and Zachary Dodds. 2025. Adaptable Metrics to Inform Introductory CS. CCSC-SW 2025, March 28-29, 2025, San Diego, CA. [**Best Poster Runner-Up**]

Harvey Mudd College Summer Research & Scholarship Poster Session | Harvey Mudd College

Sep. 2024

- Edward Donson and Sandra Brown. 2024. Development of Potential Energy Surfaces for Ionic Liquids using Artificial Neural Networks. Harvey Mudd College Summer Research and Scholarship Poster Symposium. 2024, September, 2024, Claremont, CA.

AWARDS AND HONORS

Wing and Ellen Tam Award

May 2026

- Annual award presented to Harvey Mudd College students demonstrating excellence in software design and development

Henry A. Kreiger Award

Nov. 2025

- Annual award presented to Harvey Mudd College seniors who shows particular promise in probability, statistics, or operations research

Dean's List

Jan. 2024 - Aug. 2025

- Maintained 3.0 or above while taking 15 credits or more per semester

LEADERSHIP

Harvey Mudd Applied Math Club | Co-President

Aug. 2024 – Mar. 2026

- Coordinate with SIAM, INFORMS and HMC Student Government to allocate upwards of \$2000 to activities and talks related to Applied Math.
- Organize informational panels on math competitions, resulting in record registration of students in the Mathematical Competition in Modeling
- Schedule well-attended talks, up to 75, where students can learn of applied math related careers and network with industry professionals.

Mudders Exploring Careers in Tech | Mentor

Aug. 2024 – May 2025

- Introduced mentee to on-campus and online resources for getting involved in computer science and software engineering
- Advocated for and recommended mentee to join positions of research and club leadership

PROJECTS AND ORGANIZATIONS

Hyperschedule | *React, CSS, Git, UI Design*

Nov. 2023 – Aug. 2025

- Improved UX/UI experience by implementing advanced cursor highlights to schedules using React and CSS
- Assisted in negotiating \$1,000 in funding from Harvey Mudd College and based on server and database requirements

Kanji Trainer | *Flutter, Dart, Git, Data Processing, Node.js, RESTful API*

Nov. 2022 – May 2023

- Engineered a data collection front-end software, which successfully collected thousands of handwritten Kanji characters, using Flutter software development kit
- Tested Kanji drawing software, resulting in a 100% back-end request success rate, utilizing Node.js and RESTful API convention